

****Technical Feasibility****

****KPI 1: Construction Schedule Adherence Rate****

****Definition:**** Percentage of key construction milestones (e.g., track laying, station structural completion, signaling installation) completed on or ahead of the baseline schedule. This measures the reliability of the project's delivery timeline.

****Measurement Method:**** Track planned vs. actual completion dates for each major milestone using the project management information system (e.g., Primavera or MS Project). Calculate: $(\text{Number of milestones completed on/ahead of schedule} \div \text{Total milestones}) \times 100$. Update monthly from site reports and schedule baseline.

****Rationale:**** On-time delivery of construction and technology packages is critical for technical feasibility. Delays in urban rail projects often cascade into cost overruns and interface conflicts (civil, systems, rolling stock). This KPI provides early warning of schedule risk and supports proactive mitigation.

****KPI 2: Systems Integration Readiness Index****

****Definition:**** Percentage of critical system interfaces (signaling, power supply, communications, rolling stock, and station systems) that have successfully completed factory acceptance tests (FAT), site acceptance tests (SAT), and interface verification tests according to the integration plan.

****Measurement Method:**** Maintain a systems integration matrix listing all critical interfaces. Score each as "Ready" only after documented successful test completion and sign-off. Calculate: $(\text{Number of interfaces verified ready} \div \text{Total critical interfaces}) \times 100$. Review bi-weekly during the systems phase.

****Rationale:**** Urban rail success depends on seamless integration of multiple technical subsystems. Measuring readiness of interfaces rather than individual components ensures the overall system can operate safely and reliably when the line opens.

****Environmental Sustainability****

****KPI 1: Embodied Carbon Intensity of Construction****

****Definition:**** Total greenhouse gas emissions (in tonnes CO₂e) associated with materials, construction activities, and temporary works, normalized per kilometre of new track constructed.

****Measurement Method:**** Use a project carbon calculator (aligned with recognized standards such as PAS 2080 or equivalent) fed by material quantities, supplier EPDs, and fuel/energy data from contractors. Calculate cumulative embodied carbon ÷ track kilometres completed. Report quarterly.

****Rationale:**** Construction-phase emissions are a major environmental impact of rail expansion. Tracking intensity drives material choices (low-carbon concrete/steel), construction methods, and supplier selection, directly supporting Metroville's climate goals.

****KPI 2: Modal Shift Contribution to City Green Targets****

****Definition:**** Estimated annual reduction in passenger-car vehicle-kilometres travelled (VKT) attributable to the new rail service, expressed as a percentage of Metroville's official transport-related emissions reduction target.

****Measurement Method:**** Combine ridership forecasts (or actual post-opening data) with mode-shift surveys and average trip lengths. Convert shifted trips into avoided car VKT using standard emission factors. Compare against the city's published annual transport GHG reduction goal. Update annually (forecast during construction; actual after opening).

****Rationale:**** The long-term environmental value of urban rail lies in shifting trips from private cars. Linking project performance to the city's formal green targets demonstrates strategic alignment and quantifies the contribution beyond the project boundary.

****Community Acceptance****

****KPI 1: Community Engagement Satisfaction Score****

****Definition:**** Average satisfaction rating (on a 1–5 scale) from residents, businesses, and community groups who participated in project consultations,

information sessions, or feedback channels regarding how well their input was considered and how effectively the project team communicated.

****Measurement Method:**** Administer standardized post-engagement surveys (online and paper) after major consultation events and through ongoing feedback portals. Calculate the mean score across all respondents. Segment by stakeholder type and geography. Report after each major engagement round and quarterly overall.

****Rationale:**** Perceived fairness and responsiveness of the engagement process strongly influence public acceptance. A measurable satisfaction score highlights strengths and gaps early, reducing the risk of organized opposition or delays caused by community concerns.

****KPI 2: Accessibility Improvement Index for Priority Populations****

****Definition:**** Percentage improvement in average journey time and number of barrier-free access points for residents in designated equity priority zones (low-income, high-senior, or mobility-impaired neighbourhoods) compared with the pre-project baseline.

****Measurement Method:**** Use GIS travel-time analysis (before and after network modelling) plus on-site audits of step-free access, elevators, and tactile wayfinding at new/ upgraded stations serving priority zones. Calculate weighted average improvement in travel time and access features. Report at design freeze and again at service opening.

****Rationale:**** Community acceptance is strengthened when the project demonstrably improves daily mobility for underserved groups. Measuring tangible accessibility gains for priority populations shows equitable benefits and supports long-term social licence to operate.